

MANGAIA, Cook Islands

WMO: 918480

Lat: 21.917S

Lon: 157.950W

Elev: 135

StdP: 14.62

Time Zone: -10.00 (W10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	60.9	63.0	50.0	53.6	67.7	51.7	57.2	67.5	26.9	69.8	22.3	71.3	3.4	120	0.266

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.9	85.8	77.4	84.6	77.2	83.7	76.9	79.8	83.3	78.9	82.5	78.2	81.9	8.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.2	82.5	77.7	144.9	81.4	77.2	142.1	80.9	43.5	83.2	42.6	82.9	41.8	82.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.2	15.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.9	78.2	79.5	79.0	76.8	74.5	72.4	71.2	71.2	71.3	73.1	75.5	77.0
	DBStd	3.87	2.03	1.79	1.96	2.27	2.69	2.65	2.98	2.93	2.80	2.49	2.27	2.39
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	0	0	1	0	0	0	0
	CDD50	9101	874	826	899	803	759	671	657	657	638	716	764	837
	CDD65	3628	409	406	434	353	294	221	193	193	188	251	314	372
	CDH74	19064	2975	3448	3464	2020	1140	488	372	322	328	697	1478	2332
	CDH80	2143	367	593	542	149	41	13	8	8	5	23	132	262
Wind	WSAvg	6.8	6.5	6.9	5.9	7.2	6.7	6.3	7.3	7.3	6.9	6.6	6.8	6.8
Precipitation	PrecAvg	77.5	9.9	8.8	7.1	9.6	6.7	4.0	5.0	4.6	4.1	5.2	5.5	6.8
	PrecMax	107.0	24.3	19.8	14.9	20.9	18.2	13.7	16.6	12.6	9.4	13.8	16.7	20.7
	PrecMin	43.0	0.9	0.9	1.2	1.1	0.6	0.6	1.0	0.4	0.8	0.7	0.8	0.9
	PrecStd	17.7	5.8	4.5	4.1	5.4	4.4	3.1	3.6	3.5	2.4	3.3	3.9	4.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	87.9	86.6	84.5	82.6	81.7	80.8	80.6	80.9	82.3	86.0	85.7
		MCWB	78.2	76.1	78.8	77.7	75.8	73.4	72.6	72.7	72.7	72.9	76.4	77.1
	2%	DB	84.3	86.1	85.3	83.0	81.2	79.2	79.0	78.3	78.9	80.4	82.9	84.0
		MCWB	77.5	77.3	78.2	77.0	74.9	73.1	72.9	72.1	72.6	72.3	74.7	76.7
	5%	DB	83.2	84.7	84.2	81.7	80.0	78.0	77.6	77.1	77.4	79.0	81.4	82.6
		MCWB	76.8	77.2	77.7	76.1	74.0	72.2	71.8	71.5	71.3	71.8	73.9	75.8
	10%	DB	82.2	83.6	83.1	80.7	78.8	76.7	76.3	75.8	76.0	77.5	79.9	81.3
		MCWB	76.2	76.9	77.0	75.2	73.1	71.0	70.8	70.7	70.3	71.1	73.3	75.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.1	80.7	81.1	79.9	77.7	76.6	75.6	74.9	75.5	76.2	78.0	79.0
		MCDB	84.7	84.3	84.9	82.6	80.9	79.0	78.5	77.7	78.7	79.5	82.6	82.9
	2%	WB	79.1	79.3	79.7	78.6	76.5	75.2	74.4	74.1	74.2	74.8	76.8	78.0
		MCDB	82.2	83.0	83.7	81.3	79.8	78.4	77.5	76.9	77.3	78.1	81.0	82.1
	5%	WB	78.2	78.5	78.8	77.7	75.6	73.8	73.2	73.0	73.0	73.7	75.8	77.2
		MCDB	81.4	82.3	82.8	80.7	78.6	77.1	76.3	75.8	76.1	77.2	79.7	81.1
	10%	WB	77.4	78.0	78.1	76.5	74.6	72.5	72.1	72.1	71.8	72.7	74.7	76.4
		MCDB	80.8	81.9	81.9	79.7	77.8	76.0	75.2	75.0	75.1	76.3	78.5	80.1
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	8.1	6.9	7.9	8.4	8.6	8.2	8.5	8.5	8.3	7.9
		MCDBR	8.7	9.5	9.3	8.1	8.8	9.6	10.3	9.8	9.8	10.1	10.7	9.5
		MCWBR	4.5	3.9	4.5	4.6	5.0	5.9	6.0	6.0	5.8	5.4	5.0	4.8
	5% WB	MCDBR	7.8	7.3	8.1	6.8	7.3	7.8	7.8	7.5	8.7	8.3	9.0	8.4
		MCWBR	4.9	4.0	4.7	4.8	5.2	6.2	5.9	6.3	6.3	5.8	5.3	4.8
Clear-Sky Solar Irradiance	taub	0.380	0.380	0.372	0.376	0.367	0.358	0.355	0.355	0.372	0.368	0.371	0.377	
	taud	2.619	2.624	2.655	2.633	2.639	2.643	2.639	2.632	2.578	2.587	2.608	2.611	
	Ebn at Noon	306	302	296	282	272	269	274	285	292	302	306	307	
	Edh at Noon	33	32	30	28	26	25	26	28	31	33	33	33	
All-Sky Solar Radiation	RadAvg	2034	1947	1757	1473	1193	1107	1177	1400	1679	1902	2011	2017	
	RadStd	161	159	138	90	97	80	82	82	99	134	152	168	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page